Indian Institute of Science Education and Research Mohali

IDC216 : Statistics for Scientists (2025-26) **Task 04 (October 24, 2025)**

Warm up discussion

Problem 1. Marine biologists hypothesize that a new algae strain absorbs more CO_2 than existing species. Describe the steps for setting up a statistical test and explain the roles of H_0, H_a, test statistic, α, p-value, Type I and Type II errors.

Solution

1. **Formulate hypotheses:**

$$H_0 : \mu_{\text{new}} = \mu_{\text{existing}} \quad \text{(no increase)}$$

$$H_a : \mu_{\text{new}} > \mu_{\text{existing}} \quad \text{(new strain absorbs more)}$$

   Here, the alternative hypothesis has directionality. Hence we will use one-sided test.

2. **Choose significance level α.** Typical default $\alpha = 0.05$; select smaller α (e.g. 0.01) if false positives are costly.

3. **Collect data with an appropriate design.** Ensure independence, random sampling, and sample size that minimizes β.

4. **Compute test statistic.** For comparing two means, use suitable t-statistic. Call it t_{obs}.

5. **Compute p-value.** Under H_0, calculate probability of seeing a statistic as extreme or more than t_{obs} (in the direction of H_a).

6. **Decision rule.** If $p < \alpha$, reject H_0 (evidence for H_a); otherwise fail to reject. The condition $p < \alpha$ is same as $t_{\text{df},\alpha} < t_{\text{obs}}$.

7. **Report results with context.** Give test statistic, degrees of freedom, p-value, confidence interval for effect size, and domain interpretation.

8. **Consider errors:**

   - *Type I error (probability = α):* rejecting H_0 when it is true (false claim that the new strain is better).
   - *Type II error (probability = β):* failing to reject H_0 when H_a is true (missing a real improvement).

9. **Address power.** The quantity $1 - \beta$ is called *power of the test*. Perform power calculations before study to choose sample size that keeps β small.

Remember

Hypothesis testing formalizes the question: "*How surprising is your data if the null hypothesis were true?*" The p-value quantifies surprise; α is the self-imposed threshold for acceptable surprise.

Problem 2. A materials scientist obtains $p = 0.18$ upon testing the null hypothesis $H_0 : \mu = \mu_0$, where μ is the mean tensile strength of a material.

1. What does $p = 0.18$ mean?
2. Should she reject at $\alpha = 0.05$?
3. Why does a large p not prove H_0?

Solution

1. **Meaning of $p = 0.18$:** If the null hypothesis were true (e.g., the tensile strength equals the hypothesised value μ_0), observing data as extreme as (or more extreme than) what has been observed, would happen with probability 0.18 (18%). This is not rare under H_0.
2. **Decision at $\alpha = 0.05$:** Since $p = 0.18 > 0.05$, we *fail to reject* H_0.
3. **Why large p does not prove H_0:**
   - A large p indicates the data is *compatible* with H_0, not that H_0 is true.
   - Absence of evidence is not evidence of absence: the study may have low power (small n, large variability) and thus might miss a real but small effect (Type II error).

Questions.

1. In a physics laboratory, two groups of students independently measured the specific heat capacity of a new alloy using the same calorimeter method.

Group	n	$\bar{x}$ (J/g K)	s (J/g K)
A	9	0.392	0.012
B	8	0.401	0.010

   Assume that the population variances of the two groups are equal and that the data come from approximately normal distributions.

   (a) State the null and alternative hypotheses to test whether the mean specific heat capacities of the two groups differ.

   (b) Compute the pooled variance and the t statistic.

   (c) Determine the degrees of freedom and the critical value of t at $\alpha = 0.05$ (two-tailed).

(d) Conclude whether the observed difference is significant.

2. Two different catalysts were tested to check if they yield the same average reaction rate for a certain chemical process.

Catalyst	Sample size	Standard deviation
A	12	2.4
B	10	2.0

The mean reaction rates (in mol/min) were 48.5 and 45.9 for catalysts A and B respectively. Assuming the population variances are equal, perform a pooled t-test to determine if the catalysts differ significantly in mean rate.

3. Ten patients had their blood sugar levels measured before and after a 4-week treatment using a new herbal extract. The differences (Before *minus* After) in mg/dL were:

$$[12, 8, 10, 15, 9, 7, 11, 10, 12, 8].$$

Examine whether the treatment significantly lowers blood sugar on average at the 1% level of significance. Write the hypotheses, compute the t statistic, and interpret the result.

4. The course IDC216 aims to improve students' Statistical ability. To evaluate the impact of the course, the same 10 students were given a test *before* and *after* completing IDC216. Their scores (out of 20) are as follows:

Student	Before IDC216	After IDC216
1	10	15
2	12	14
3	9	13
4	11	15
5	13	16
6	14	17
7	8	13
8	11	14
9	10	12
10	9	13

(a) Define the null and alternative hypotheses.

(b) Compute the mean and standard deviation of the paired differences.

(c) Calculate the t statistic and degrees of freedom.

(d) At $\alpha = 0.05$, decide whether the course significantly improved student performance.